PROJECT DOCUMENTATION

Formula 1 Race Intelligence Dashboard

Data Analytics • Tableau • Flask Web Integration

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Project Documentation Phase

1. INTRODUCTION

1.1 Project Overview

Formula 1 Race Intelligence Dashboard is a data-visualization project that presents selected Formula 1 racing information in an interactive Tableau dashboard and a companion Tableau Story. The dashboard brings together race-record coverage, driver and team performance, tyre-compound usage, podium achievements, reliability-related retirement reasons, race-hosting locations, and season-based comparisons. A Flask-based website provides a central entry point to the published dashboard and story. The source code is hosted on GitHub and the website is deployed on Render.

1.2 Purpose

The purpose is to make Formula 1 information easier to explore than reviewing individual records separately. Users can compare drivers and teams, inspect tyre usage, explore retirement reasons, and use dashboard filters to investigate the data. The project also demonstrates the integration of a Tableau analytics product with a small Python web application and public deployment.

Project Links

Live website: https://f1-race-intelligence-mhgm.onrender.com/

GitHub repository: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

Tableau dashboard:
https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

Tableau Story: https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence

2. IDEATION PHASE

2.1 Problem Statement

Formula 1 data contains multiple dimensions, including drivers, teams, seasons, circuits, tyre compounds, race outcomes, and retirement reasons. Without a consolidated visual interface, users may find it difficult to compare performance or identify where to investigate further. The project addresses this by presenting selected metrics and comparisons in an interactive dashboard, supported by a guided story and a web page that makes the published views easy to access.

2.2 Empathy Map Canvas

Primary user: a Formula 1 enthusiast, student, or analyst who wants to explore racing data without writing queries or manually comparing many records.

- Says: “I want to compare drivers and teams and understand patterns across seasons.”
- Thinks: “A single dashboard should make important performance measures easier to compare.”
- Does: Selects filters, compares chart marks, reviews driver/team rankings, and navigates the Tableau Story.
- Feels: More confident exploring the data when labels, charts, and filters are organized in one place.
- Pain points: Multiple data fields, difficulty comparing many records, and uncertainty when chart axes or aggregations are unclear.
- Needs: Readable visualizations, clear labels, relevant filters, a guided story, and dependable access to the published project.

2.3 Brainstorming

Ideas considered for the solution included:

- Create KPI cards for the scope of the available dataset.
- Use a driver leaderboard and team ranking for quick performance comparisons.
- Show tyre-compound distribution using a pie chart.
- Include podium achievements and reliability/DNF-related views.
- Use a map to show race-hosting locations.
- Provide season-based views for team performance and tyre usage.
- Add filters so users can investigate a subset of seasons, teams, tyre compounds, or DNF reasons.
- Create a Tableau Story to present selected views in a guided sequence.
- Build a Flask website that links to the dashboard and story.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Journey stage	User action	Requirement / opportunity
Discover	User opens the live project website.	Clear project title and visible navigation.
Choose a view	User selects Dashboard or Story.	Working links to Tableau Public views.
Explore	User reads KPI cards and chart titles.	Readable labels, legends, and useful chart layout.
Filter	User selects season, team, tyre compound, or DNF reason.	Filters should update the applicable views.
Interpret	User compares driver/team results and other measures.	Consistent metrics and clear axis/aggregation definitions.
Share	User shares the public site or repository.	Public URLs and source-code access.

3.2 Solution Requirement

- Functional: display Formula 1 dashboard and story through publicly accessible links.
- Functional: show race-record and season/circuit coverage indicators visible in the dashboard.
- Functional: allow exploration of driver/team performance, tyre usage, podiums, reliability/DNF reasons, and race-hosting geography.
- Functional: provide available filters for season, team, tyre compound, and DNF reason.
- Functional: offer a website entry point implemented with Flask.
- Non-functional: readable visualization labels and a usable page layout.
- Non-functional: public access to the web application and Tableau views.
- Non-functional: source code available in GitHub for review and maintenance.

3.3 Data Flow Diagram

High-level data and interaction flow:

User / Viewer → Render-hosted Flask Website → Tableau Public Dashboard or Story

The Tableau workbook uses the Formula 1 dataset configured in the workbook. Tableau Public serves the published visualizations. The Flask application provides the project landing page and access to the published views. GitHub stores the application source and related project files.

Note: the exact dataset source URL was not present in the supplied project evidence and should be added from the actual source used; it is not guessed in this report.

3.4 Technology Stack

Technology	Role in project
Tableau Public	Interactive dashboard, charts, filters, and Tableau Story.
Python	Application programming language for the web component.
Flask	Web framework serving the project page.
HTML/CSS	Page structure and presentation.
Git/GitHub	Version control and public source-code repository.
Render	Hosting platform for the live web application.
Gunicorn	WSGI server used by the deployment configuration.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The dashboard is aligned with the problem because it places several types of Formula 1 analysis in one interface and supports interactive filtering. The Tableau Story supports a guided presentation of selected information, while the Flask website gives viewers a simple starting point. The solution is suitable for exploratory analysis and demonstration; any chart with an unclear axis, aggregation, or null indicator should be validated before using it for a strong quantitative conclusion.

4.2 Proposed Solution

- Publish an interactive Tableau dashboard with summary metrics and multiple analysis views.
- Provide driver and team comparison charts, tyre-compound distribution, podium and reliability views, a host-location map, and season-oriented charts.
- Include filters to let users narrow the dashboard to selected categories.
- Publish a Tableau Story for a guided narrative.
- Build and deploy a Flask website that links to the dashboard and story.
- Keep project source code available through GitHub.

4.3 Solution Architecture

- Presentation layer: browser-based project landing page hosted on Render.
- Web application layer: Flask application provides the homepage/navigation.
- Analytics layer: Tableau Public hosts the published dashboard and story.
- Source management: GitHub repository stores application files and deployment configuration.
- Viewer flow: browser → Render website → Tableau dashboard/story.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Activity	Description
1. Define objective	Set the project goal and choose Formula 1 analysis as the topic.
2. Inspect data	Review the dataset fields and identify the measures

	and dimensions relevant to the analysis.
3. Prepare data	Check field types, nulls, and consistency; only record transformations that were actually performed.
4. Build charts	Create the driver, team, tyre, podium, reliability, map, and season-based views.
5. Assemble dashboard	Arrange the charts and KPI cards and configure the available filters.
6. Build story	Arrange selected visualizations into a guided sequence.
7. Publish Tableau views	Publish the dashboard and story to Tableau Public.
8. Develop web page	Create the Flask landing page and add navigation to the published Tableau views.
9. Deploy	Connect the GitHub repository to Render and deploy the application.
10. Verify	Open the live website, dashboard, story, and repository; check access and basic navigation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project evidence confirms that a Tableau dashboard and story have been published and a Flask website is deployed. The supplied evidence does not include measured page-load times, dashboard render times, filter-response timings, concurrent-user results, or a formal load-test report. Accordingly, this report records the verifiable checks and does not invent performance measurements.

Test	Method	Evidence/status	Result / note
Live website access	Open the Render URL in a browser.	Live deployment URL is provided; re-check at submission time.	Access/availability check; no timing value supplied.
Dashboard access	Open the Tableau Public dashboard URL.	Published dashboard link is provided.	Confirm charts render and filters respond.
Story access	Open the Tableau Story URL.	Published Story link is provided.	Confirm story points navigate as expected.
Navigation	Use Dashboard and Story links from the website.	Website is intended as the central entry point.	Record any broken links if found.
Filter interaction	Change season, team, tyre compound, and DNF reason filters.	Filters are visible in the supplied dashboard screenshot.	Measure response time only if tested.
Data/chart validation	Check axis labels, aggregation, null indicators, and filter scope.	Some chart views need validation before strong interpretation.	Review completion-rate and tyre-usage season charts; inspect podium null indicator.
Load/performance	Measure initial load and interactions using a repeatable method.	No measurements supplied.	Not quantified in this report.

7. RESULTS

The supplied dashboard screenshot displays 6,464 race records, 17 seasons analysed, and 24 circuits represented. It includes a driver performance leaderboard, tyre-compound usage, team performance ranking, podium achievements, driver reliability/DNF reasons, a host-country map, a completion-rate view, season-wise team championship performance, and tyre usage by season.

7.1 Output Screenshots

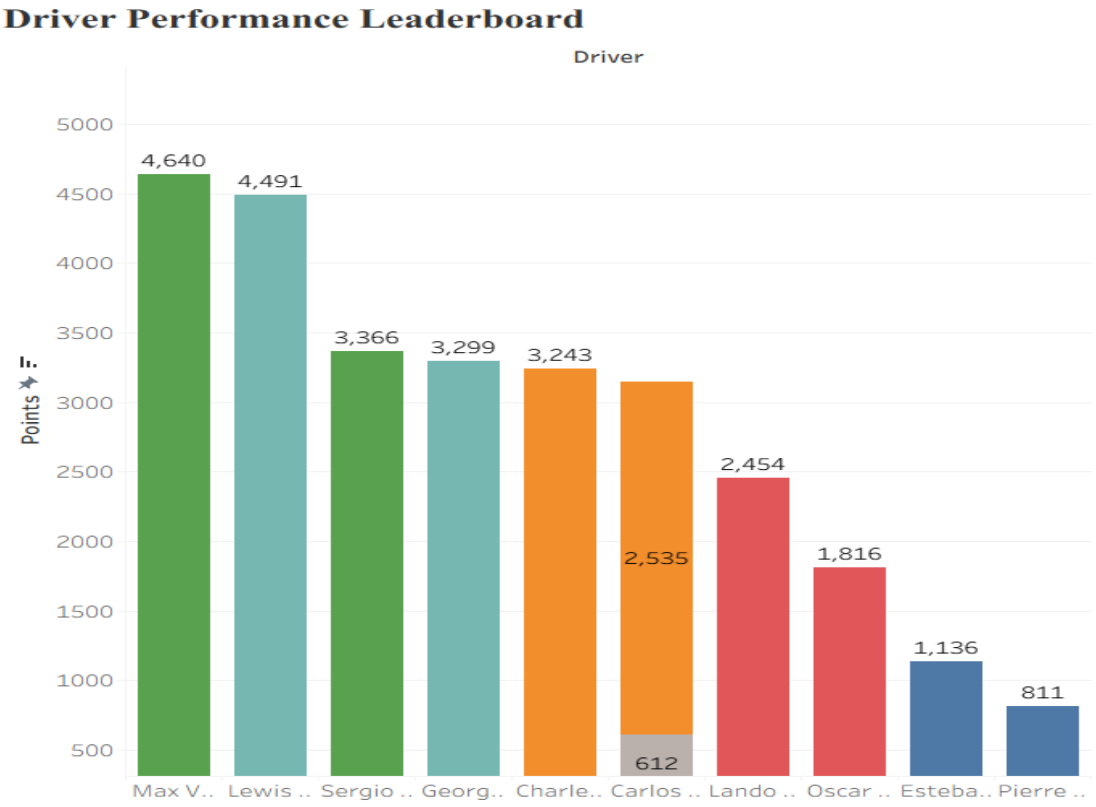


Figure 2. Driver Performance Leaderboard.

Tyre Compound Usage

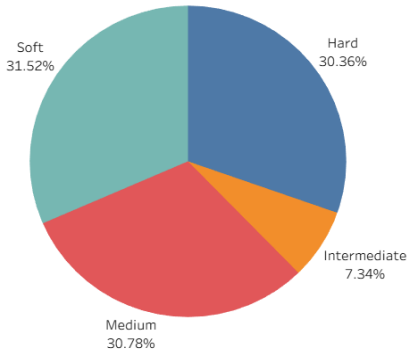


Figure 3. Tyre Compound Usage.

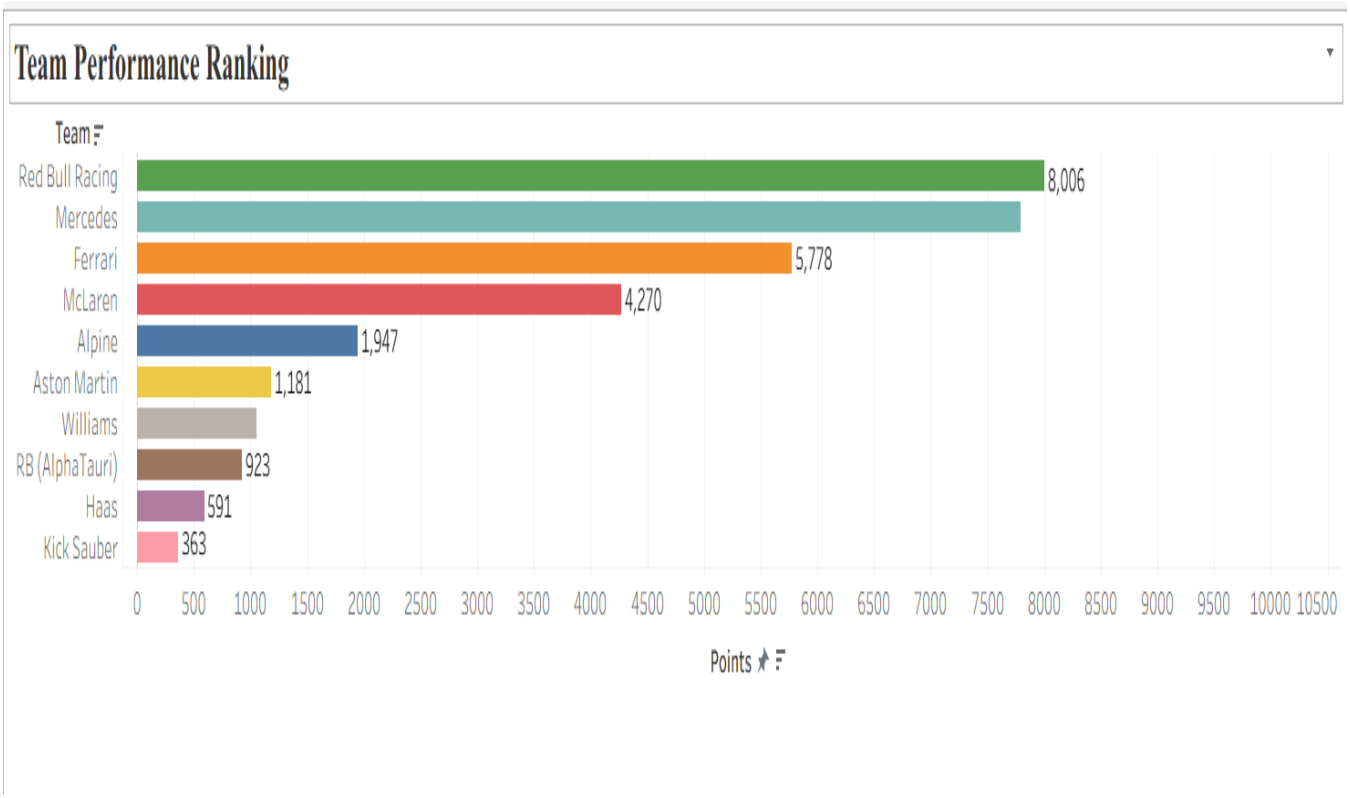


Figure 4. Team Performance Ranking.

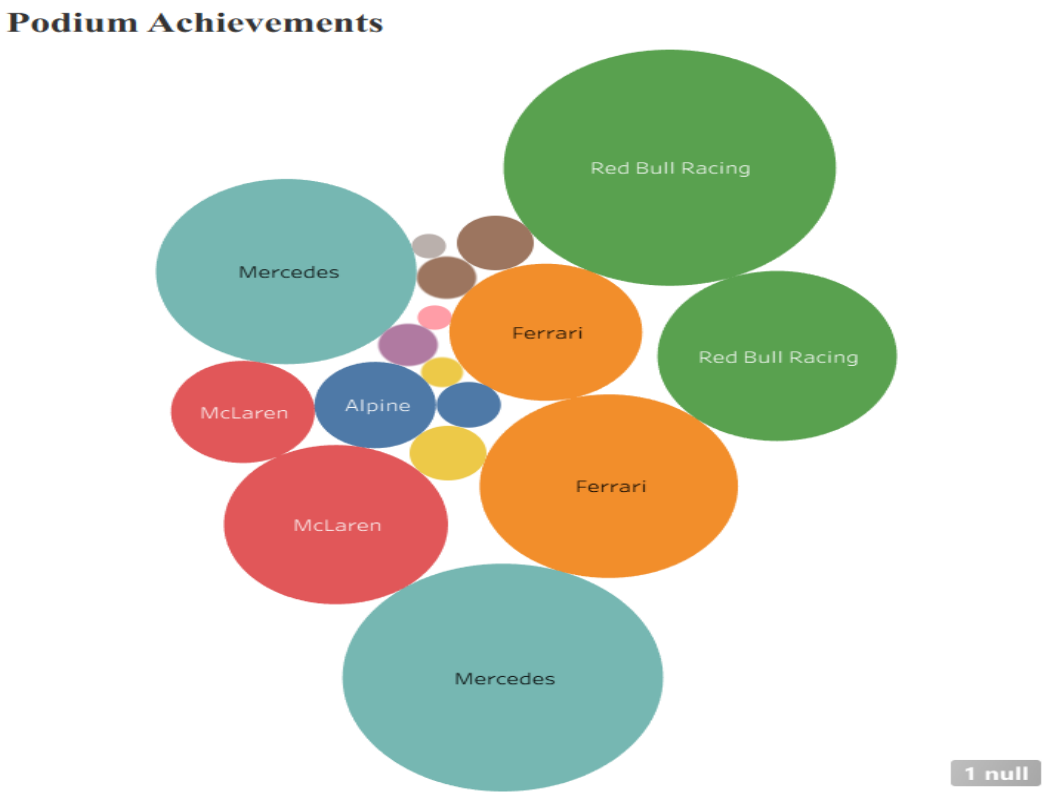


Figure 5. Podium Achievements. The dashboard shows a null indicator; validate before reporting exact values.

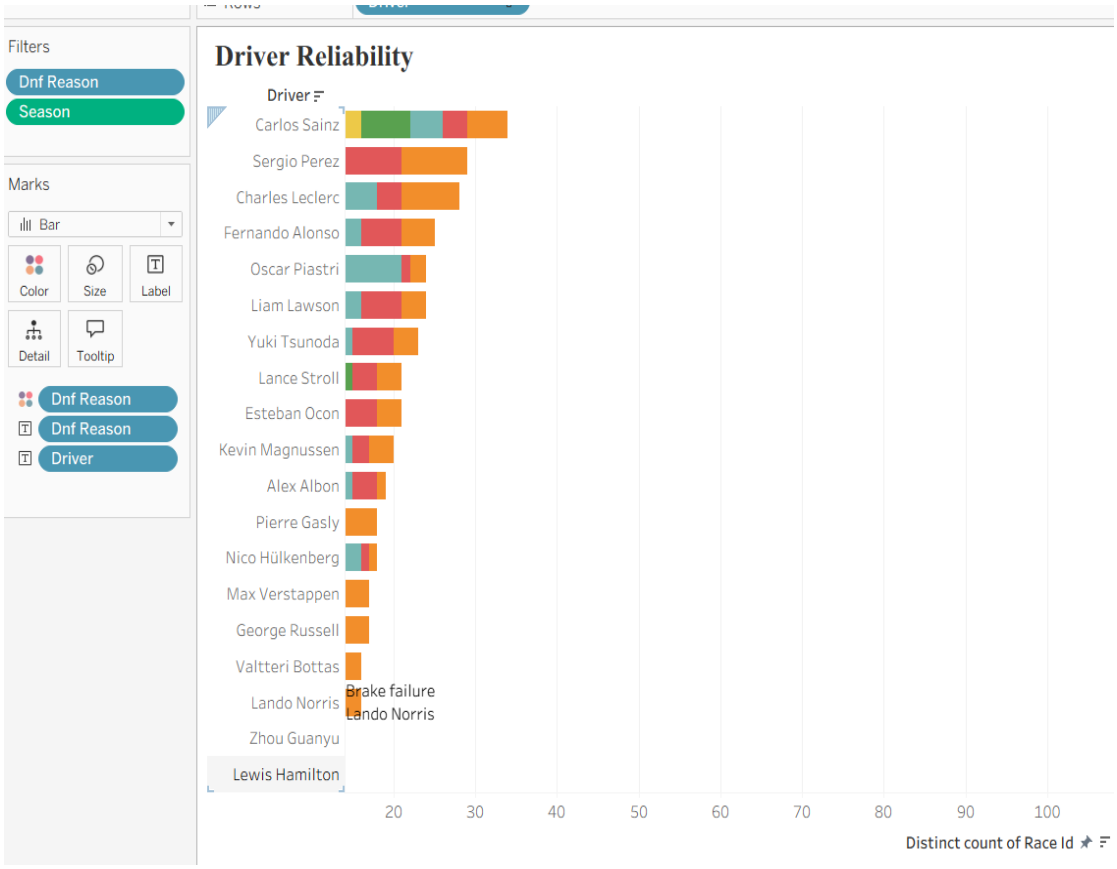


Figure 6. Driver Reliability and DNF Reason exploration.

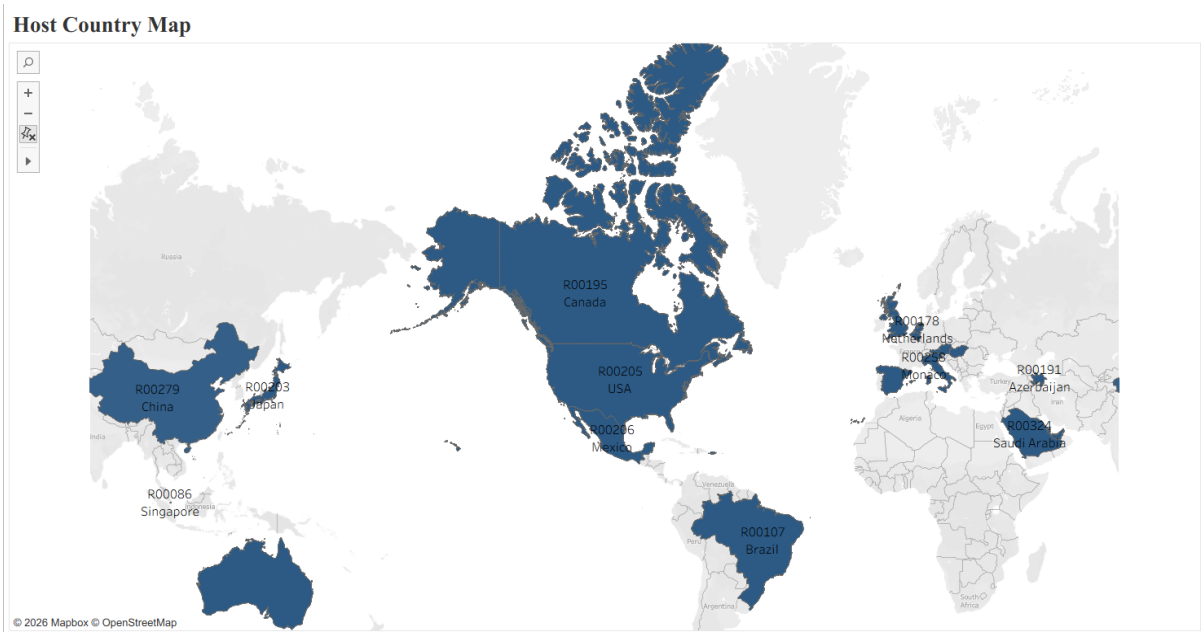


Figure 7. Host Country Map; represents race-host locations.

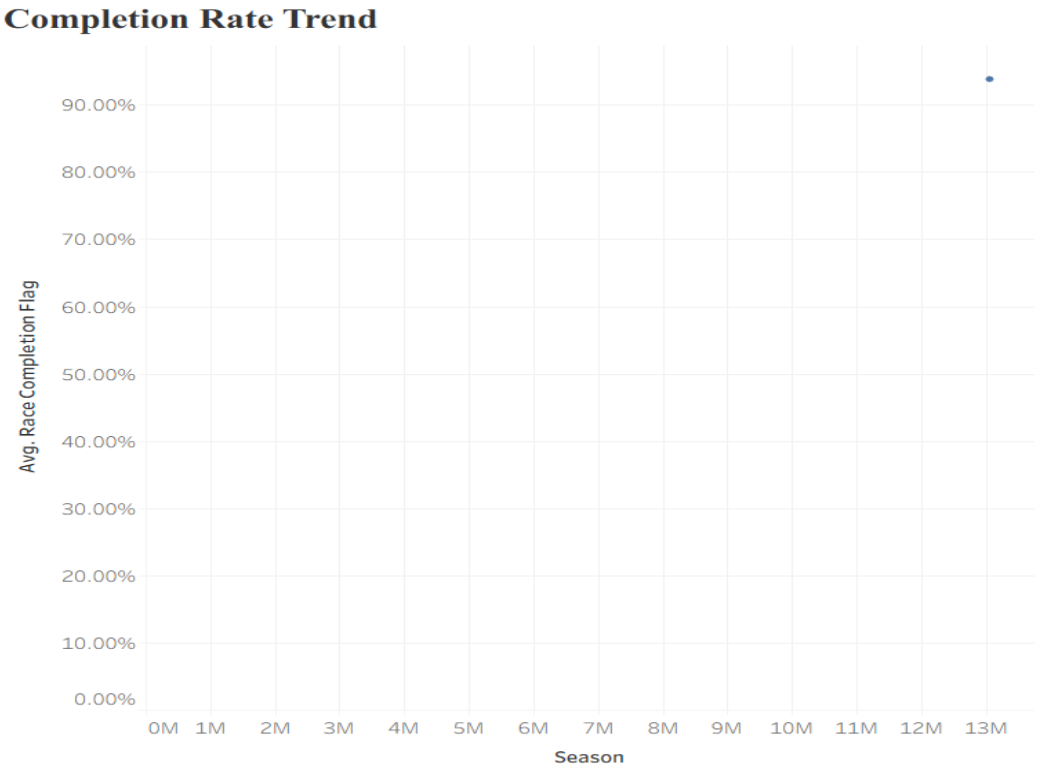


Figure 8. Completion Rate Trend. Validate the axis and aggregation before interpreting a trend.

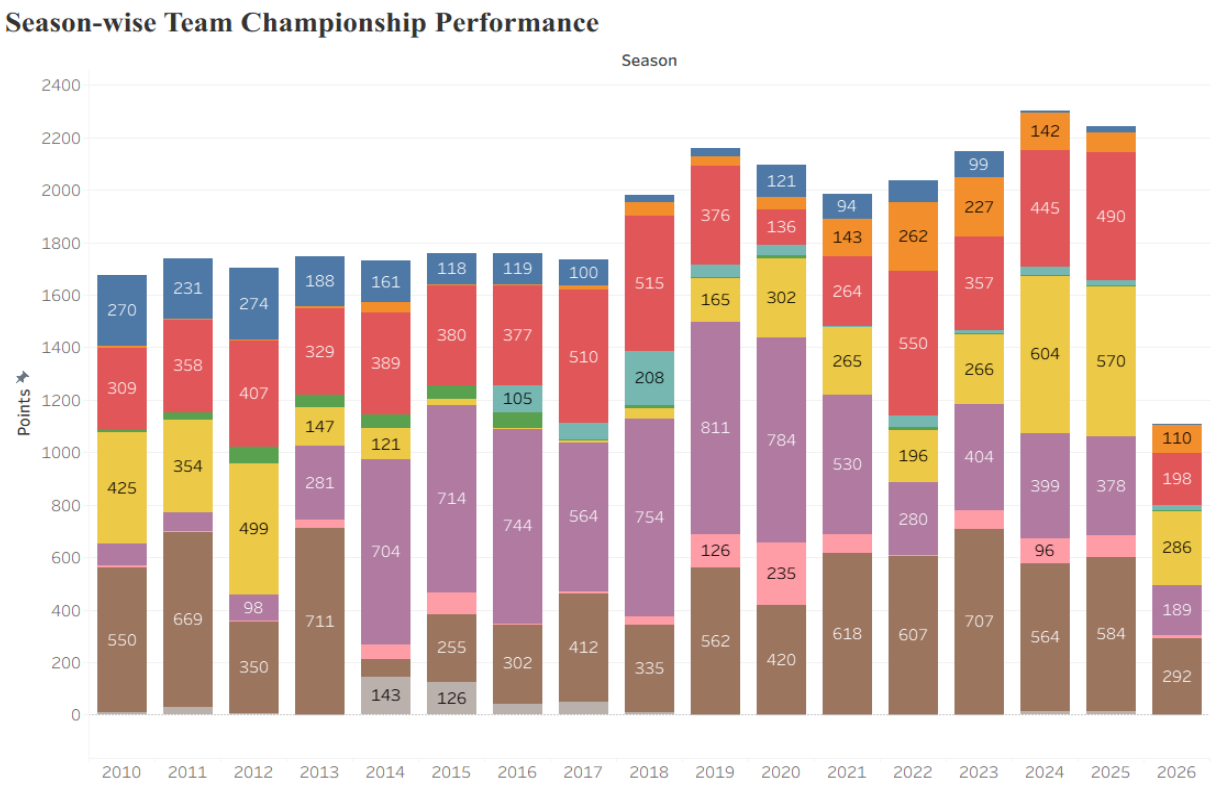
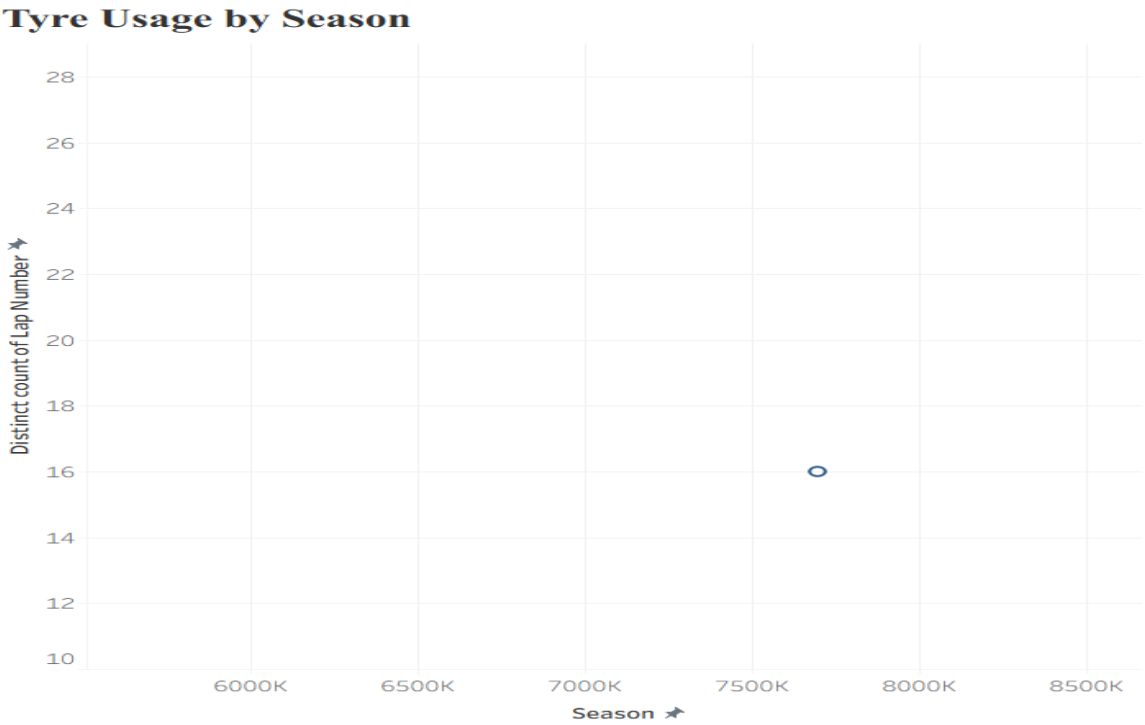


Figure 9. Season-wise Team Championship Performance.

Figure 10. Tyre Usage by Season. Validate the measure and aggregation.



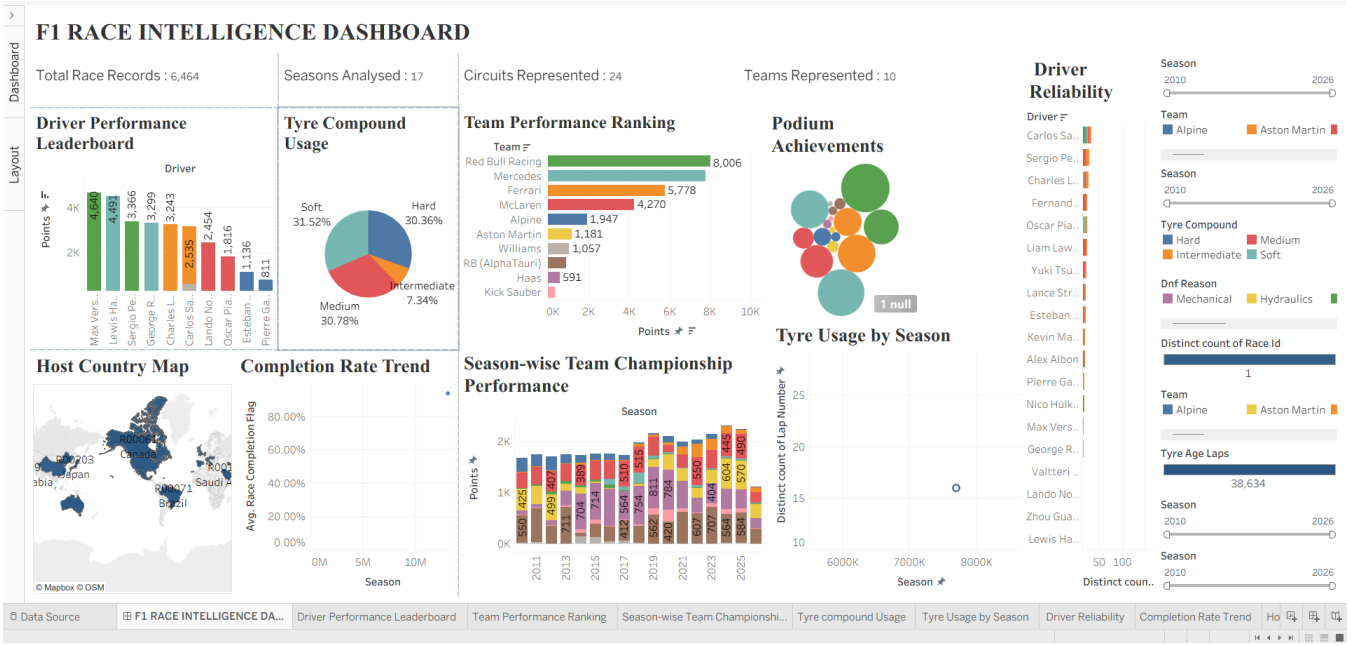


Figure 11. Dashboard filters.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines several Formula 1 analysis views in a single dashboard.
- Provides quick overview indicators for record, season, and circuit coverage.
- Enables driver and team comparisons through charts.
- Includes filters for interactive exploration.
- Provides a Tableau Story for guided communication.
- Offers a web entry point and publicly accessible source repository.

Disadvantages / Limitations

- Interpretation depends on the accuracy, completeness, and date coverage of the underlying dataset.
- The exact dataset source and its licensing/source details should be stated in the final submission.
- The supplied screenshot indicates that some charts may need axis/aggregation validation; a podium view also shows a null indicator.
- No measured performance or load-test results were supplied, so quantitative performance claims cannot be made.
- The free-hosting environment may have a cold-start delay after inactivity; this should be considered during a live demonstration.

9. CONCLUSION

The Formula 1 Race Intelligence Dashboard demonstrates a complete analytics presentation flow using Tableau Public and a companion Flask website. It brings together driver and team comparisons, tyre-compound distribution, podium and reliability exploration, geographic context, and season-oriented views. The published Tableau Story supports guided exploration, while the Render deployment and GitHub repository make the project accessible for demonstration and review. Before final submission, the dataset source should be documented and the charts with unclear aggregations or null values should be checked in Tableau.

10. FUTURE SCOPE

- Add the exact dataset source, version/date range, and data dictionary to the documentation.
- Validate chart calculations, axis fields, aggregation levels, and null values.
- Add clear definitions for points, podiums, DNF reasons, tyre age, and completion rate.
- Run and document repeatable performance checks for website loading, dashboard rendering, and filter response.
- Improve accessibility with descriptive chart text, readable labels, keyboard-friendly navigation, and suitable contrast.
- Add a user-friendly loading or error message when an external Tableau view is slow or unavailable.
- Expand the analysis with additional seasons or metrics only when the underlying dataset supports them.

11. APPENDIX

Source Code (if any)

GitHub repository: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

The repository is the reference location for the project source. This report does not reproduce source code that has not been independently checked against the repository contents.

Dataset Link

The dataset source URL was not identifiable from the evidence supplied for this report. Insert the exact original dataset URL and any required attribution here before submission. Do not substitute a guessed source.

GitHub & Project Demo Link

GitHub: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

Project demo website: https://f1-race-intelligence-mhgm.onrender.com/

Tableau dashboard:
https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

Tableau Story: https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence

Submission checklist: confirm the dataset source; re-open all public links; review chart calculations noted in Section 7; add any measured performance-test results only if actually tested; and confirm whether the institution requires PDF, Word, or both.